$X \to Y \to \hat{X}$

$\hat{X}_{MMSE} = E(X|Y)$ Gaussian MMSE = LMMSE

$MSE = EE(X - \hat{X}_{MMSE})^2 = ED(X|Y)$

Let $\hat{X} = g(Y)$, $MSE = EE(X - \hat{X})^2$

$\min_g MSE \to \min_{\hat{X}} E(X - \hat{X})^2 | Y$

$X^2 - 2X\hat{X} + \hat{X}^2$

$\min_{\hat{X}} \cancel{EX^2} - 2\hat{X}E(X|Y) + \hat{X}^2 \Rightarrow \hat{X}_{MMSE} = E(X|Y)$

$\hat{X}_{LMMSE} = E(X) + \frac{cov(X,Y)}{D(Y)}(Y - E(Y))$

$MSE = D(X)(1-\rho^2)$ Linear Non-Homoegeneous

$\hat{X}_{LH} = \frac{E(XY)}{E(Y^2)} \cdot Y$ Linear Homoegeneous

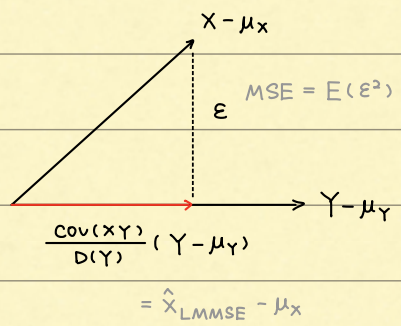
当 $EX = EY = 0$

即 $\hat{X}_{LMMSE} = \hat{X}_{LH}$

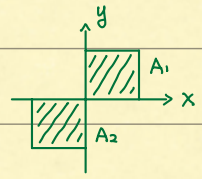
$MSE = E(X^2)(1 - \frac{E^2(XY)}{E(X^2)E(Y^2)})$

$\hat{X}_{LH} = aY$, $MSE = E(X - aY)^2$

$\min_a EX^2 + a^2EY^2 - 2aE(XY)$ $\hat{a} = \frac{E(XY)}{E(Y)^2}$



e.g.




Uniform in A, Find $\hat{X}_{MMSE}$ and MSE

Find $\hat{X}_{LMMSE}$ and MSE

$$\hat{X}_{MMSE} = E(X|Y) = \begin{cases} 0.5, & 0 \le y \le 1 \\ -0.5, & -1 \le y \le 0 \end{cases}$$

$$MSE = E(D(X|Y))$$

1°: In $A_1$, $D(X|Y) = \frac{1}{12}$, $A_2$ same

so $MSE = \frac{1}{12}$

2°: $\iint_A (x - \hat{X}_{MMSE})^2 P_{XY}\, dx dy = 2\iint_{A_1} (x - \frac{1}{2})^2 \cdot \frac{1}{2}\, dx dy = \int_0^1 (x - \frac{1}{2})^2 dx = \frac{1}{12}$

$$\hat{X}_{LMMSE} = \cancel{E(X)} + \frac{cov(X,Y)}{D(Y)}(Y - \cancel{E(Y)})$$

$Y \sim U[-1,1]$, $D(Y) = \frac{1}{12} \times 2^2 = \frac{1}{3}$ $\quad cov(X,Y) = E(XY) - \cancel{E(X)}E(Y)$

$E(XY) = \iint_A xy \cdot \frac{1}{2}\, dx dy = 2\iint_{A_1} xy \cdot \frac{1}{2}\, dx dy = \frac{1}{4}$

$\hat{X}_{LMMSE} = \frac{3}{4}Y$ （并非 $\hat{X}_{LMMSE} = Y$）

$MSE = D(X)(1-\rho^2) = \frac{1}{3} \times (1 - \frac{9}{16}) = \frac{7}{48}$ $\quad \rho = \frac{cov(X,Y)}{\sqrt{DX} \cdot \sqrt{DY}} = \frac{\frac{1}{4}}{\frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}}} = \frac{3}{4}$